Independent Written Research and Advisory Memorandum

The NAGMO Investment Thesis: What It Says, What It Proves, and What an Investor Should Watch

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A plain-language memorandum for an informed general reader, including readers encountering portfolio theory for the first time.

Executive summary

The NAGMO Investment Thesis is a proposed way to invest in the artificial-intelligence (AI) industry by betting that one part of the industry may do better than two other parts. In its simplest form, it is long NVIDIA and short Amazon and Google, while assigning no active position to Meta. The central economic idea is understandable: companies that sell the equipment needed to build AI systems may benefit differently from companies that spend money building and selling AI services.

The research papers develop this idea into a much wider framework. They examine whether the portfolio is genuinely market-neutral, how interest-rate conditions affect the assessment, what safer and more aggressive versions might look like, how borrowing and liquidity can create losses, and how competition between AI companies may include unusually aggressive financial behaviour.

My overall assessment is as follows:

1. **The investment idea is coherent, but it is a hypothesis, not a proven result.** NVIDIA may outperform Amazon and Google in some futures; it may underperform in others. The outcome depends on AI demand, pricing power, competition, capital spending, and the ability of each company to turn AI activity into lasting profits.

2. **The original portfolio is arithmetically balanced under its stated inputs.** That is not the same as being protected from a general stock-market fall. A balanced set of dollar amounts can still rise or fall with the market, technology shares, interest rates, or AI-sector sentiment.

3. **The number 3.42 is not automatically a leverage ratio.** It is the sum of the absolute values of the original portfolio coefficients. It becomes a conventional measure of gross exposure only after the positions are translated into a consistent, clearly defined capital base.

4. **Short positions create special risks.** A short seller can be forced to buy back shares at a much higher price, may face rising borrowing costs, and can suffer losses that are not capped at the original amount invested.

5. **The inflation-risk-premium argument is a useful warning about assumptions, not a licence to declare the portfolio safe.** The papers show how a critique can change when the assumed monetary environment changes. Their numerical turning point is illustrative, not a verified live-market threshold.

6. **AI-sector valuation must include strategic resilience.** A company can have strong technology and still be vulnerable if financing, suppliers, customers, collateral, or market confidence come under pressure. The adversarial-attack papers provide theoretical models of such risks; they do not prove that any named company has carried out an attack.

The practical recommendation is to treat NAGMO as a research framework that may support a carefully controlled relative-value investment, not as a ready-made market-neutral trade. Before real capital is committed, the investor should define the actual position sizes, measure market and factor exposures, verify short availability and financing costs, set loss limits, and test the thesis across several plausible AI and monetary scenarios.

1 What is NAGMO trying to do?

Imagine that you believe two kinds of businesses will benefit from AI, but not equally. One group sells the essential tools used to build AI systems: processors, accelerators, networking equipment, and related infrastructure. Another group operates cloud platforms and sells services that use AI. The NAGMO thesis asks whether the first group—represented by NVIDIA—might perform better than two large businesses with important cloud and service operations—Amazon and Google.

In investment language, *long* means buying an asset because you expect its price to rise. *Short* means borrowing and selling an asset because you expect its price to fall, or at least to rise less than the asset on the long side. A relative-value trade can make money when the long investment performs better than the short investments, even if the whole market is rising or falling. But that outcome is not guaranteed.

The original NAGMO construction uses the following positions:

Company	Original coefficient
NVIDIA	+1 (long)
Amazon	−1 (short)
Google	−1 (short)
Meta	0 (no active position)
Balancing component	−0.42

The original paper uses market-capitalisation figures of USD 5.03 trillion for NVIDIA, USD 2.40 trillion for Amazon, USD 2.21 trillion for Google, and USD 1.24 trillion for Meta. These are the inputs used in that paper, not current market quotations. With the stated coefficients, the balancing calculation is:

$$5.03 - 2.40 - 2.21 - 0.42 = 0. \tag{1}$$

This shows that the chosen numbers add to zero under the paper's particular scaling. It does not show that the portfolio has no exposure to the stock market or to the technology sector.

2 The most important distinction: dollar balance is not market neutrality

Suppose a seesaw has two children of equal weight, one on each side. The seesaw may balance. But if a strong wind pushes both children in the same direction, the seesaw can still move. Portfolio balance works in a similar way.

A dollar-balanced portfolio has offsetting dollar amounts according to a chosen calculation. A market-neutral portfolio is designed so that broad market movements have little effect on its returns. These are different goals.

A share price does not move only because of the overall market. It also moves because of company-specific news and shared influences, such as interest rates, technology valuations, semiconductor demand, cloud spending, and investor enthusiasm for AI. NVIDIA, Amazon, and Google can react differently to the same event. Their price movements therefore do not automatically cancel just because the initial dollar amounts balance.

A common measure of market sensitivity is *beta*. In simple terms, beta estimates how much a share or portfolio tends to move when the broad market moves. A portfolio with a beta near zero has historically shown little sensitivity to that market measure over the estimation period. But beta can change, and a zero market beta would still not remove every other risk.

The papers are therefore right to insist on a distinction between an accounting identity and measured risk. The original arithmetic is valid on its own terms; the stronger claim of market neutrality needs additional evidence and a clear definition of the positions.

3 What does the figure 3.42 mean?

The original paper reports a coefficient measure called W. It adds the absolute values of the portfolio coefficients:

$$W = |1| + |-1| + |-1| + |0| + |-0.42| = 3.42. \tag{2}$$

This is a valid calculation. It tells us the total size of the listed coefficients after taking away their signs. It does not, by itself, tell us how much money must be borrowed, how much capital is at risk, or how much leverage an investor is using.

To calculate conventional gross exposure, an investor must first state the actual market value of every position and the amount of investor capital supporting those positions. The paper's coefficient vector also combines dimensionless position coefficients with a balancing component described in monetary terms. Those units must be made consistent before the figure can be treated as a real-world leverage measure.

In plain English: 3.42 is a property of the paper's mathematical construction. It should not be advertised as "3.42 times leverage" unless the portfolio has been translated into a fully specified account with consistent units.

4 Interest rates and the inflation risk premium: why assumptions matter

The Conditional Impartiality paper focuses on the US inflation risk premium, written as ϕ. This is the extra return investors may demand for bearing uncertainty about future inflation. It is one part of the explanation for nominal bond yields.

The paper's central point is that judgements about beta, financing costs, and the relative sensitivity of growth companies to interest rates depend on the monetary environment. If an analyst assumes one inflation-risk-premium regime and never tests another, the analyst may mistake a conditional conclusion for a universal one.

The paper uses an illustrative model in which the estimated portfolio beta changes with ϕ, and it identifies a turning point around 0.5 percentage points under its chosen calibration. That number should be read carefully. It is generated by a stylised model and illustrative parameter choices; it is not presented here as a live, independently verified market threshold.

There is another important distinction. Showing that a verdict changes under different assumptions is a useful sensitivity test. It does not prove that every criticism is wrong, or that the portfolio is safe when the inflation risk premium becomes negative. The right lesson is that an evaluation should disclose its assumptions and test more than one plausible monetary regime.

An investor should therefore ask: what happens if inflation compensation rises, falls, or becomes negative? What happens if real interest rates rise? What happens if financing costs increase while the short positions become harder to maintain? Those questions should be answered with current data before a trade is sized.

5 The portfolio choices: cautious, balanced, and aggressive

The portfolio-construction paper does not claim that one set of weights is best for every investor. It lays out a family of possible designs. This is sensible because people have different tolerances for losses, different time horizons, and different abilities to manage short positions.

Approach	Plain-language aim	Main trade-off
Low-gross construction	Use smaller positions so that a mistake is less damaging.	Lower potential gains as well as lower exposure.
Constrained beta-neutral spread	Choose the long and short sizes to reduce broad-market sensitivity.	Depends on estimated betas, which can change.
Multi-factor controlled portfolio	Limit exposure to several shared drivers, not just the overall market.	Requires more data and more complicated monitoring.
Scenario-limited defensive portfolio	Set a maximum acceptable loss under named stress situations.	The chosen scenarios may miss a new kind of shock.
Conviction-weighted construction	Make larger positions where the investor has stronger evidence.	A strong belief can still be wrong.

Controlled high-gross construction	Take a larger relative-value position within a defined risk budget.	Financing, short squeezes, and losses can grow quickly.
Regime-aware construction	Adjust exposure when monetary or AI-sector conditions change.	Signals can be noisy and adjustments can be costly.

The papers deliberately do not supply supposedly optimal numerical weights. That restraint is important. To calculate defensible weights, one would need reliable return histories, correlations, factor exposures, borrow availability, borrow fees, financing rates, trading costs, and a well-defined risk budget. A mathematical formula cannot manufacture missing evidence.

For a cautious investor, smaller gross exposure, explicit loss limits, and a simpler structure are generally easier to supervise. A more aggressive investor may choose larger positions, but should not confuse a larger expected gain with a better risk-adjusted investment.

6 Leverage, short-selling, and liquidity: how a good idea can still lose money

Leverage means using borrowed money or other financing to control a larger position than the investor's own capital would otherwise allow. Leverage magnifies outcomes. It can magnify gains, but it also magnifies losses and can trigger demands for more cash.

Short-selling has a special danger. If an investor buys a share, the share price can fall to zero, so the loss is limited to the amount invested. If an investor shorts a share, the price can rise far above the original sale price. The investor may have to buy it back at that higher price. In theory, the loss on an uncovered short position has no fixed upper limit.

Liquidity means how easily an investor can buy or sell without moving the price too much. A company may have shares that trade frequently, while the particular short position may still be difficult or expensive to borrow. Liquidity can also disappear when many investors try to exit at once.

For NAGMO, these risks matter on both sides of the trade. NVIDIA may fall sharply; Amazon or Google may rise sharply; the cost of borrowing the shorted shares may increase; a broker may demand more collateral; or the investor may be forced to close positions during a market disruption. The portfolio can be right about long-term business prospects and still lose money because it cannot survive the path taken by prices and financing conditions.

The market-risk paper makes another useful point: a large, famous company is not automatically a risk-free or always-liquid short. The relevant question is not only whether the shares normally trade in large volumes. It is whether the investor can maintain, finance, and close the actual position under stressed conditions.

7 AI competition and balance-sheet adversarial risk

The AI industry is not simply a contest to invent the best model. Companies also compete over chips, data centres, electricity, cloud contracts, skilled workers, customers, financing, and investor confidence. Their balance sheets—the assets, debts, and ownership claims shown in their accounts—can shape how long they can keep investing and how well they can withstand a shock.

The two companion studies on balance-sheet adversarial attacks explore a more severe possibility: a financial contest in which one firm takes actions that damage another firm's financial position or share price, and the other firm retaliates. Their models discuss borrowing, equity transactions, market impact, liquidation costs, and feedback between share prices and financing constraints. A second model studies how such interactions may settle into peaceful or retaliatory patterns under specified stochastic assumptions.

These are theoretical models. They help identify mechanisms worth monitoring; they are not proof that NVIDIA, Amazon, Google, Meta, or any other named company has carried out such an attack. Ordinary competition—for example, building a better product, lowering prices, investing in capacity, or signing a customer contract—must not be confused with manipulative or abusive market conduct.

Why should an investor care about the theoretical possibility? Because a share-price fall can sometimes cause more than a temporary loss on a screen. If a company relies on its share price or securities as collateral, a fall may make borrowing harder. If borrowing becomes harder, the company may delay

investment or sell assets. That may then affect suppliers, customers, and future profits. The initial market shock can become a real business problem.

The reverse is also possible: a sharp share-price fall may have little effect on the company's actual products, customers, cash generation, or long-term competitive position. A sound valuation must distinguish a temporary market disturbance from genuine damage to the business.

The valuation paper's most important contribution is therefore to ask two questions separately: *What is the business worth if it can operate normally?* and *How resilient is the business if a strategic financial contest or market disruption occurs?*

8 How should AI companies be valued?

A company's value should ultimately depend on the cash it can generate for its owners over time, adjusted for risk and the time value of money. In everyday terms, a business is not worth a great deal merely because it is exciting. It must eventually turn its technology, customers, and market position into durable profits and cash.

The AI-sector valuation paper recommends examining four layers:

1. **Productive capability:** Does the company have useful technology, computing capacity, infrastructure, talent, and reliable products?
2. **Commercial conversion:** Can it turn those capabilities into paying customers, recurring revenue, and sustainable profit margins?
3. **Financial resilience:** Can it fund investment, meet debt obligations, and withstand a period when expected revenue arrives late or costs rise?
4. **Strategic and market-structure resilience:** Can it withstand supplier dependence, customer concentration, competitive retaliation, market disruption, regulation, and geopolitical restrictions?

This approach prevents a common mistake: counting the same economic benefit several times. For example, a cloud provider may buy AI chips, use them in a data centre, and sell computing services to customers. The chip maker's revenue and the cloud provider's revenue are connected. An analyst should not simply add every company's AI-related revenue as though each dollar represented a separate dollar of final demand. The analyst must identify who pays, who earns the margin, and where the economic value is ultimately retained.

The papers also distinguish *fundamental value* from *market price*. Fundamental value is an estimate of the business's underlying economic worth. Market price is the price at which shares are currently traded. They can differ. A market panic can push price below a reasonable estimate of value; excessive optimism can push price above it. Neither difference should be assumed permanent without evidence.

9 The scenarios that could change the answer

The valuation framework does not rely on a single prediction about the future. It asks what happens under several different futures. The following scenarios are qualitative stress tests, not assigned probabilities or price targets.

Scenario	What it would mean for the investment thesis
AI infrastructure supercycle	Demand for computing capacity stays very strong. Companies supplying essential hardware and infrastructure may capture substantial profits. This could support the long NVIDIA side, although the share price must still be compared with the profits expected.
Cloud and service monetisation catch-up	Cloud and AI service providers turn heavy investment into strong recurring revenue and profits. Amazon and Google could outperform the hardware supplier, weakening the original NAGMO spread.

Compute becomes cheaper and more abundant	More efficient models, alternative chips, or better software reduce the amount of expensive computing needed. This could reduce the scarcity premium enjoyed by hardware suppliers.
Capital-spending overbuild	Companies build too many data centres or buy too much equipment relative to customer demand. Revenue may disappoint while depreciation, power, and financing costs remain high.
Financing and liquidity shock	Interest rates, credit spreads, collateral demands, or share-borrow costs rise. Even a fundamentally sound relative-value view may become difficult or expensive to maintain.
Adversarial escalation	Strategic financial actions, market-impact events, or retaliation create price and funding pressure. The effect may remain temporary or may spread into investment, customers, and operations.
Regulatory or geopolitical fracture	Export controls, energy constraints, data rules, or international tensions disrupt suppliers, markets, or customers.
Broad AI disappointment	Adoption, willingness to pay, or productivity gains fall short of expectations. Valuations across the sector may decline together, and the relative winner may still deliver a negative absolute return.

A useful investor question is not simply "Which company will win?" It is "What would have to be true for my investment to make money, and what evidence would show that I am wrong?" For NAGMO, the investor must identify the conditions under which NVIDIA outperforms both Amazon and Google by enough to cover financing, borrow, and trading costs.

10 What would make the NAGMO thesis stronger or weaker?

The thesis would become stronger if evidence showed that demand for AI infrastructure remains durable, NVIDIA retains pricing power and a defensible competitive position, and the cloud and service businesses struggle to convert their AI spending into comparable profits. It would also become stronger if the actual portfolio could be financed and maintained at reasonable cost, with acceptable market and factor exposures.

The thesis would become weaker if custom chips or competing accelerators reduce NVIDIA's margins, if AI services generate strong and profitable revenue for Amazon or Google, if customers reduce infrastructure spending, or if the original relative-performance gap is already more than reflected in market prices. It would also become weaker if short borrow becomes expensive or unreliable, if factor exposures are larger than the investor can tolerate, or if the portfolio cannot survive a plausible temporary drawdown.

A crucial point: being correct about the industry does not guarantee making money from the shares. If NVIDIA is an excellent company but its stock price already assumes extraordinary future success, the investment may still disappoint. Likewise, Amazon or Google can be strong businesses and still be attractive shorts only if their share prices and expected returns justify that view. Business quality and investment attractiveness are related, but they are not identical.

11 Independent advisory recommendations

Based on the combined research, I recommend the following decision process. These are research and risk-management recommendations, not a personal instruction to buy or sell any security.

1. **Describe the trade honestly.** Until market and factor exposures are measured, call it a dollar-balanced AI-sector relative-value proposal, not a proven market-neutral portfolio.

2. **Specify real positions.** State the number of shares or monetary value of each long and short position, the account capital, the cash or debt balance, and the source of financing. Use consistent units throughout.

3. **Measure the risks that matter.** Estimate market beta and relevant factor exposures, including technology growth, semiconductor demand, cloud spending, interest rates, and AI-sector sentiment. Show the estimation period and explain that exposures can change.

4. **Price the short side before entering.** Confirm that shares can be borrowed, record the borrow fee and financing rate, understand margin rules, and model the cost of a short squeeze or forced buy-in.

5. **Choose a risk architecture before choosing weights.** Decide whether the priority is low gross exposure, beta control, multi-factor control, scenario loss limits, or a more aggressive conviction trade. Do not select weights merely because they make the opening equation look elegant.

6. **Use several monetary regimes.** Reassess the portfolio under higher, lower, and negative inflation-risk-premium assumptions, and under changes in real interest rates and financing spreads. Treat the paper's illustrative 0.5 percentage-point turning point as a model example, not as a trading signal.

7. **Value the businesses separately.** Estimate each company's expected cash generation under several AI scenarios. Avoid double-counting revenue across chip suppliers, cloud operators, and AI-service providers.

8. **Stress the path, not only the destination.** Test sharp price gaps, rising borrow costs, reduced market depth, collateral calls, delayed revenue, and simultaneous losses in the long and short legs. Ask whether the portfolio can survive until the investment thesis has time to work.

9. **Monitor evidence, not rumours.** Watch public filings, debt maturities, capital expenditure, customer concentration, supplier dependence, unusual financing arrangements, market depth, and verified regulatory disclosures. Unusual trading or a falling share price alone does not prove an adversarial attack or unlawful conduct.

10. **Write down exit and review rules.** Define in advance what evidence invalidates the thesis, what loss level triggers a review, how much capital may be committed, and when positions must be reduced. Review the assumptions when company fundamentals or market conditions change.

12 What this research does not yet establish

A clear memorandum should also state what cannot responsibly be concluded from the papers. The repository provides a conceptual and mathematical research programme. It does not provide a current, independently verified set of portfolio weights; a complete live return and covariance dataset; executable short-borrow quotes; a broker-specific margin schedule; or a verified estimate of the probability of each AI scenario.

The beta relationships and inflation-risk-premium turning point are stylised or illustrative. The adversarial market models depend on specified assumptions about financing, price impact, firm behaviour, and stochastic dynamics. Simulated algorithmic behaviour is not proof of real-world corporate conduct. No claim is made here that NVIDIA, Amazon, Google, or Meta has engaged in a balance-sheet attack against another company.

Nor does this memorandum provide a current fair value or price target for any of the companies. The market-capitalisation inputs in the original thesis are historical inputs to that paper's construction. A live investment decision would require updated financial statements, market prices, forecasts, rates, borrow costs, and legal and regulatory review.

13 Final opinion

NAGMO begins with a simple and interesting question: could the company supplying a critical part of the AI infrastructure earn more from the AI expansion than large companies selling cloud and AI services? That is a legitimate investment question. The research programme has made the question more serious by examining the assumptions hidden inside the original portfolio, the effect of monetary conditions, alternative ways of sizing the trade, the dangers of short-selling, and the possibility that strategic financial conflict can affect market prices and corporate resilience.

The strongest conclusion is also the most measured one. The original construction balances under its stated arithmetic, but that does not establish market neutrality. The relative-value thesis may work, but it is conditional on business outcomes, valuations, financing costs, and the path of market prices. The adversarial models justify monitoring financial resilience and market structure, but they do not establish

that named firms have attacked one another. The valuation framework is valuable precisely because it keeps these questions separate.

My independent advisory view is therefore: **retain NAGMO as a structured research hypothesis; do not treat the original coefficient vector as a fully validated investable portfolio.** Advance to implementation only after the actual positions are dimensionally specified, risk exposures are measured, short-side costs are confirmed, scenario losses are bounded, and the thesis has clear conditions under which it must be reconsidered.

In one sentence: *NAGMO is a thoughtful way to express a view about who may capture the economic value of AI, but the investor must prove that the trade can survive the risks of being wrong, being early, or being forced out before the market eventually agrees.*

Source basis and research independence

This memorandum synthesises the six numbered papers currently listed in the NAGMO repository: (1) *The NAGMO Investment Thesis: A Market-Neutral Portfolio in AI*; (2) *The Conditional Impartiality Theorem*; (3) *Review of the NAGMO Investment Thesis*; (4) *Alternative Risk-averse and Risk-taking Portfolio Constructions*; (5) *Market Neutrality, Factor Exposure, Leverage and Liquidity Risk*; and (6) *AI-Sector Valuation Framework and Scenario Analysis*. The two companion papers on balance-sheet adversarial attacks, supplied alongside the repository work, are treated as supporting theoretical studies because the valuation paper draws on their mechanisms.

The word *independent* describes the analytical stance of this synthesis: it evaluates the claims and limitations of the source papers rather than simply repeating their strongest conclusions. It does not mean that this memorandum is an external audit, a regulated investment-advice engagement, or an independent empirical verification of the underlying companies or models. The memorandum is based on the contents of the identified research papers and does not claim a new live-data study.

References

[1] Ghosh, S. (2025). *The NAGMO Investment Thesis: A Market-Neutral Portfolio in AI*. Kolkata, India.

[2] Ghosh, S. (with Z.ai). (2026). *The Conditional Impartiality Theorem*. Kolkata, India.

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[4] Ghosh, S. (2026). *Alternative Risk-averse and Risk-taking Portfolio Constructions*. Kolkata, India.

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[7] Ghosh, S. (2026). *Balance-Sheet Adversarial Attacks in Equity Markets: Amplified Retaliation, Market-Impact Leverage, and an Algorithmic Framework*. Kolkata, India.

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Glossary

AI infrastructure
: The chips, computers, data centres, networks, electricity, and related equipment used to develop and operate AI systems.

Long position
: Buying an investment because you expect it to rise in value.

Short position
: Borrowing and selling an investment in the hope of buying it back later at a lower price.

Relative value
: A strategy that focuses on one investment doing better or worse than another.

Market neutrality
Designing a portfolio so that broad market movements have limited influence on its return.

Beta
A measure of how sensitive an investment has been to movements in a chosen market benchmark.

Factor exposure
Sensitivity to a common influence, such as interest rates, technology stocks, or semiconductor demand.

Gross exposure
The total size of long and short positions before subtracting one from the other.

Leverage
Using borrowing or financing to control a larger position than one's own cash alone would support.

Liquidity
How easily an asset can be bought or sold without greatly changing its price.

Collateral
Assets pledged to support a loan or trading position.

Inflation risk premium
Extra compensation investors may demand for uncertainty about future inflation.

Fundamental value
An estimate of what a business is worth based on its expected future economic performance.

Scenario analysis
Testing how an investment might behave under different plausible futures.

Market impact
The effect that a trade itself has on the price at which an asset can be bought or sold.

Adversarial risk
The risk that another participant's strategic actions worsen a firm's or investor's financial position.

The End